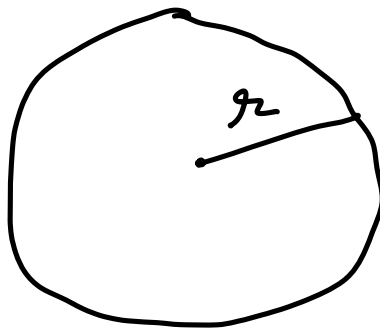
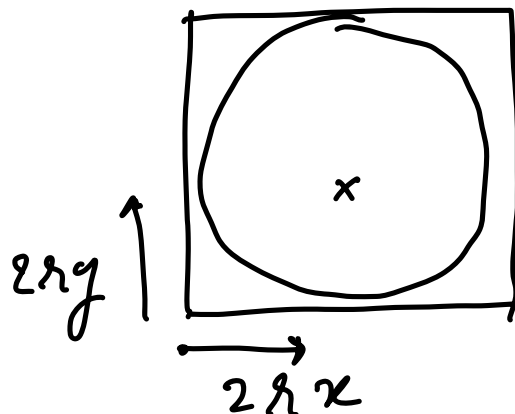


So importance sampling let's us compute $E_f[w(x)]$ but what if we want to get samples from $f(x)$ but cannot sample directly from it.

Example: Get a point x in 2D
Sampled uniformly at random from
a circle of radius r using only
 $U[0,1]$ variables



Solution:



$$x, y \in U[0,1]$$

* Sample Uniformly in square

→ Reject if outside the Circle

Now consider the generic case

$f(x)$: target distribution

$g(x)$: distribution to be used.

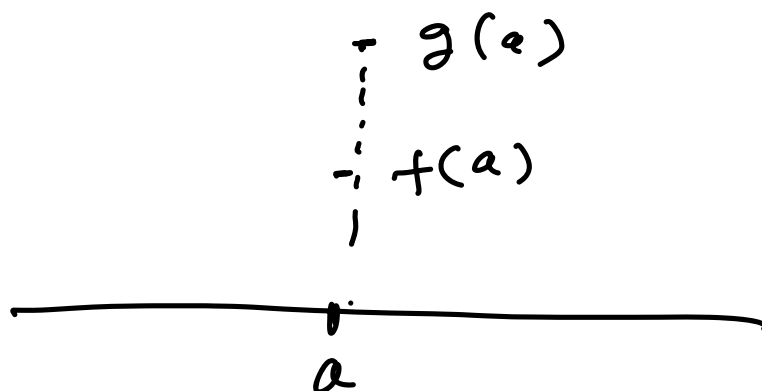
$$a \sim g(x)$$

How do we make sure that probability of getting x is actually $f(x)$?

$$\text{if } \underline{g(a) > f(a)}$$

↳ Reduce the probability.

↳ start rejecting the sample!



accept
with
prob. $\left(\frac{f(a)}{g(a)} \right)$

Algo till now

Sample $a \sim g(x)$

if $g(a) \geq f(a)$:

$r = \text{Uniform}[0,1]$

if $r > f(a)/g(a)$

Reject and go Sample again

otherwise

Accept. return a

else ??

What about if $f(a) > g(a)$

↳ we need to increase the probability of a

But how?

⇒ Decrease probability of other " x " enough so that we can relatively use $g(a)$

Envelope :=
Choose an M s.t. $Mg(x) > f(x) \forall x$

Algo:

Sample $a \sim g(x)$

$r = U[0,1]$

if $a < \frac{f(a)}{Mg(a)}$: Accept

else : Reject and Sample again

